

Quadratic functions - Set 2 - Solutions

Specification MM1.3, M4.5, M4.11-M4.14.

Answer key

Question	Answer	Topic
1	A	Quadratic functions
2	D	Quadratic functions
3	A	Quadratic functions
4	B	Quadratic functions
5	C	Quadratic functions
6	E	Quadratic functions
7	E	Quadratic functions
8	H	Quadratic functions
9	E	Quadratic functions
10	B	Quadratic functions
11	F	Quadratic functions
12	F	Quadratic functions

Worked solutions

Question 1 QUA-18 · Quadratic functions · spec MM1.3

Find the range of values of x for which $x^2 - 7x + 10 < 0$ and $x^2 - 4 > 0$.

Answer: A

$(x - 2)(x - 5) < 0$ gives $2 < x < 5$; $x^2 > 4$ gives $x < -2$ or $x > 2$. The intersection is $2 < x < 5$.

Question 2 QUA-19 · Quadratic functions · spec MM1.3

The expression $x^2 + 6x + c$ can be written as $(x + p)^2 + q$. Given that $q = -c$, find the value of c .

Answer: D

$(x + 3)^2 + (c - 9)$, so $q = c - 9 = -c$, giving $2c = 9$ and $c = \frac{9}{2}$.

Question 3 QUA-21 · Quadratic functions · spec MM1.3

The parabola $y = x^2 - 4x + 3$ is reflected in the x -axis and then translated 2 units in the positive y -direction. What is the maximum value of the resulting function?

Answer: A

$x^2 - 4x + 3 = (x - 2)^2 - 1$, minimum -1 . Reflecting gives maximum 1; translating up by 2 gives maximum 3.

Question 4 QUA-23 · Quadratic functions · spec MM1.3

The line $y = mx$ meets the curve $y = x^2 + 9$ at two distinct points. Find the complete set of values of m .

Answer: B

$x^2 - mx + 9 = 0$ needs $m^2 - 36 > 0$, so $m < -6$ or $m > 6$.

Question 5 QUA-24 · Quadratic functions · spec MM1.3

Let $f(x) = x^2 - 2x - 8$. For how many integers n is $f(n) < 0$?

Answer: C

$(x - 4)(x + 2) < 0$ gives $-2 < x < 4$; the integers are $-1, 0, 1, 2, 3$: five of them.

Question 6 X2-30 · Quadratic functions · spec MM1.3

The equation $x^2 + bx + c = 0$ has real coefficients. Which of the following is a **sufficient** condition for it to have two distinct real roots?

Answer: E

$c < 0$ makes the discriminant $b^2 - 4c$ positive. $b^2 \geq 4c$ allows equality (a repeated root); the others do not control the discriminant.

Question 7 QUA-04 · Quadratic functions · spec MM1.3

The roots of $x^2 - 5x + 2 = 0$ are α and β . Find the value of $\alpha^2 + \beta^2$.

Answer: E

$\alpha + \beta = 5$ and $\alpha\beta = 2$. So $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25 - 4 = 21$.

Question 8 QUA-06 · Quadratic functions · spec MM1.3

The function $f(x) = x^2 - 6x + 13$ is defined for all real x . Which of the following statements is/are true?

- I $f(x) > 0$ for all real x
- II The graph of $y = f(x)$ is symmetric about the line $x = 3$
- III The equation $f(x) = 4$ has exactly one real solution

Answer: H

$f(x) = (x - 3)^2 + 4 \geq 4 > 0$, so I is true. The axis of symmetry is $x = 3$, so II is true. $f(x) = 4$ only when $(x - 3)^2 = 0$, i.e. $x = 3$: exactly one solution, so III is true.

Question 9 QUA-09 · Quadratic functions · spec MM1.3

Find the sum of all real values of x satisfying $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

Answer: E

Let $u = x^2 - 3x$: $(u - 4)(u + 2) = 0$. $x^2 - 3x - 4 = 0$ gives $x = 4, -1$ (sum 3). $x^2 - 3x + 2 = 0$ gives $x = 1, 2$ (sum 3). Total 6.

Question 10 QUA-14 · Quadratic functions · spec MM1.3

The equation $x^2 - 2mx + m + 6 = 0$ has two distinct positive roots. Find the complete set of values of m .

Answer: B

Distinct real roots: $4m^2 - 4(m + 6) > 0 \Rightarrow m^2 - m - 6 > 0 \Rightarrow (m - 3)(m + 2) > 0$, so $m < -2$ or $m > 3$. Both roots positive needs sum $2m > 0$ and product $m + 6 > 0$, so $m > 0$. Combined: $m > 3$.

Question 11 QUA-17 · Quadratic functions · spec MM1.3

The quadratic equation $ax^2 + bx + c = 0$ (with $a \neq 0$) has real coefficients. Which of the following statements is/are always true?

- I If $ac < 0$, the equation has two distinct real roots.
- II If $b = 0$, the equation has no real roots.
- III If the equation has a repeated root, then $b^2 = 4ac$.

Answer: F

I: $b^2 - 4ac > 0$ whenever $ac < 0$, so true. II: $x^2 - 4 = 0$ has $b = 0$ and real roots, so false. III: a repeated root means the discriminant is zero, so true.

Question 12 QUA-20 · Quadratic functions · spec MM1.3

For how many integer values of k does $x^2 + kx + 12 = 0$ have integer roots?

Answer: F

Integer roots r, s with $rs = 12$: pairs (1, 12), (2, 6), (3, 4) and their negatives, giving $k = -(r + s) = -13, -8, -7, 13, 8, 7$: six values.